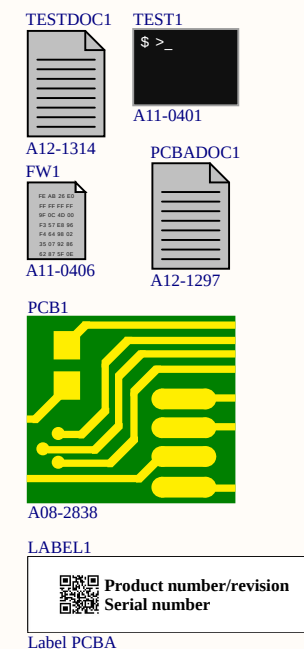
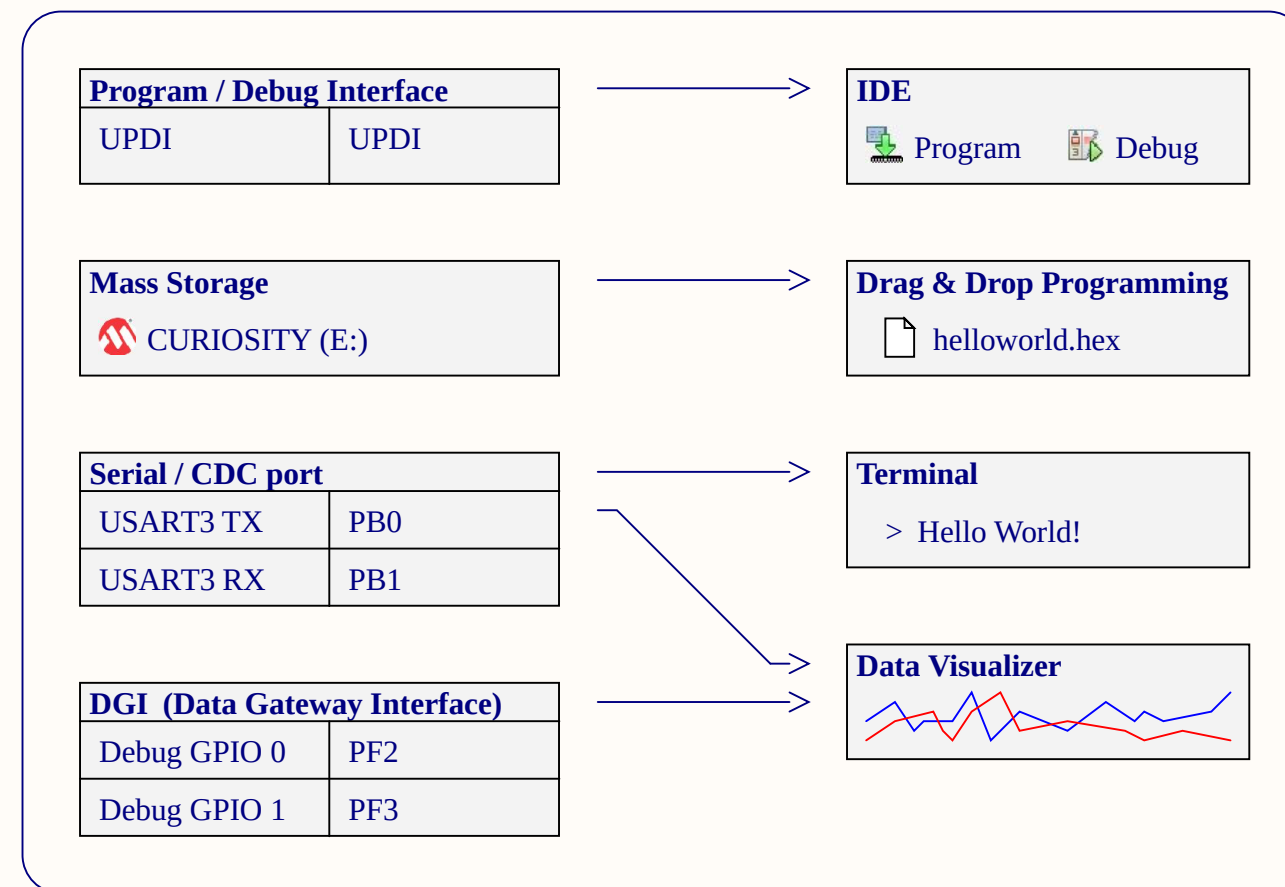
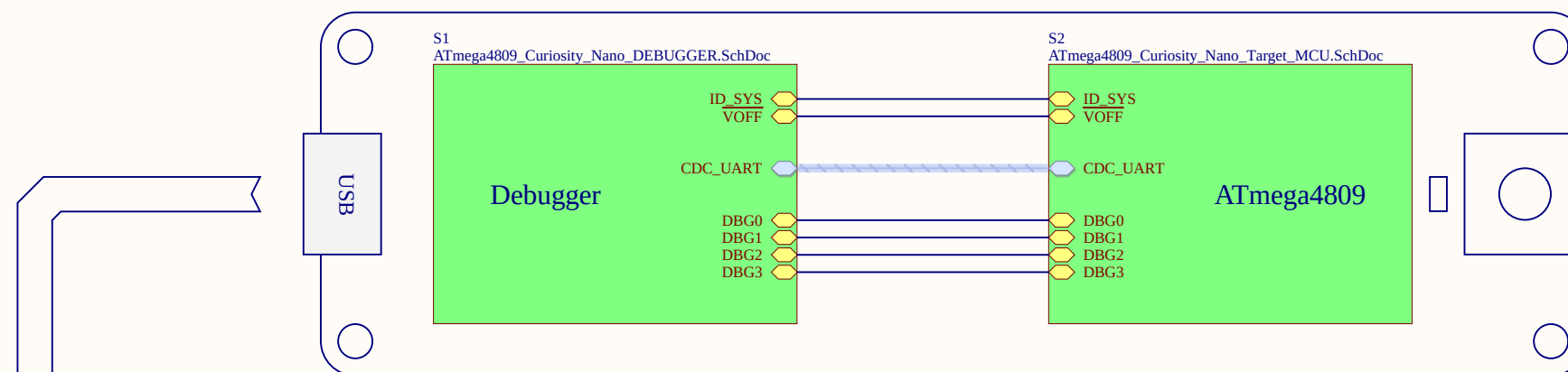


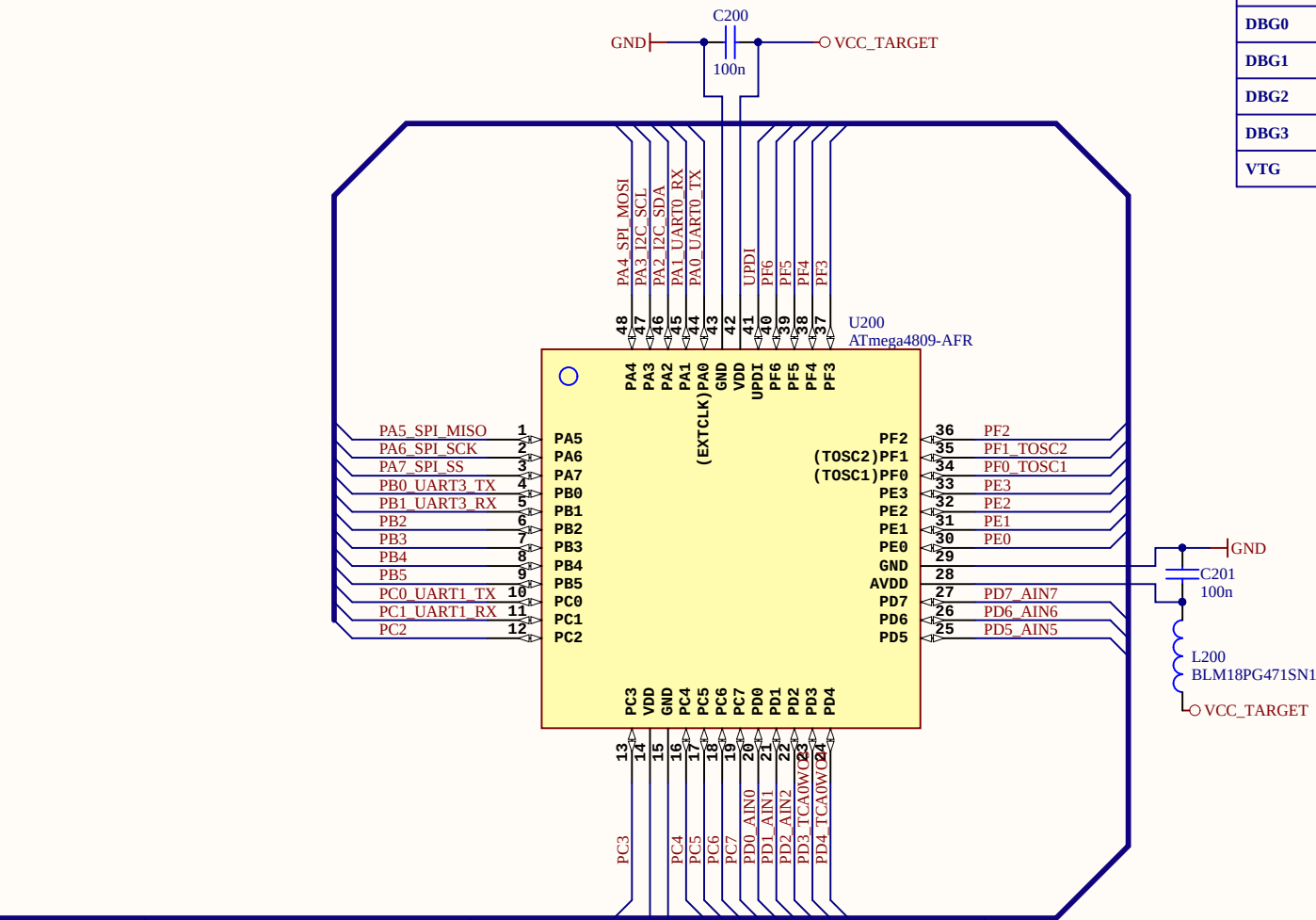
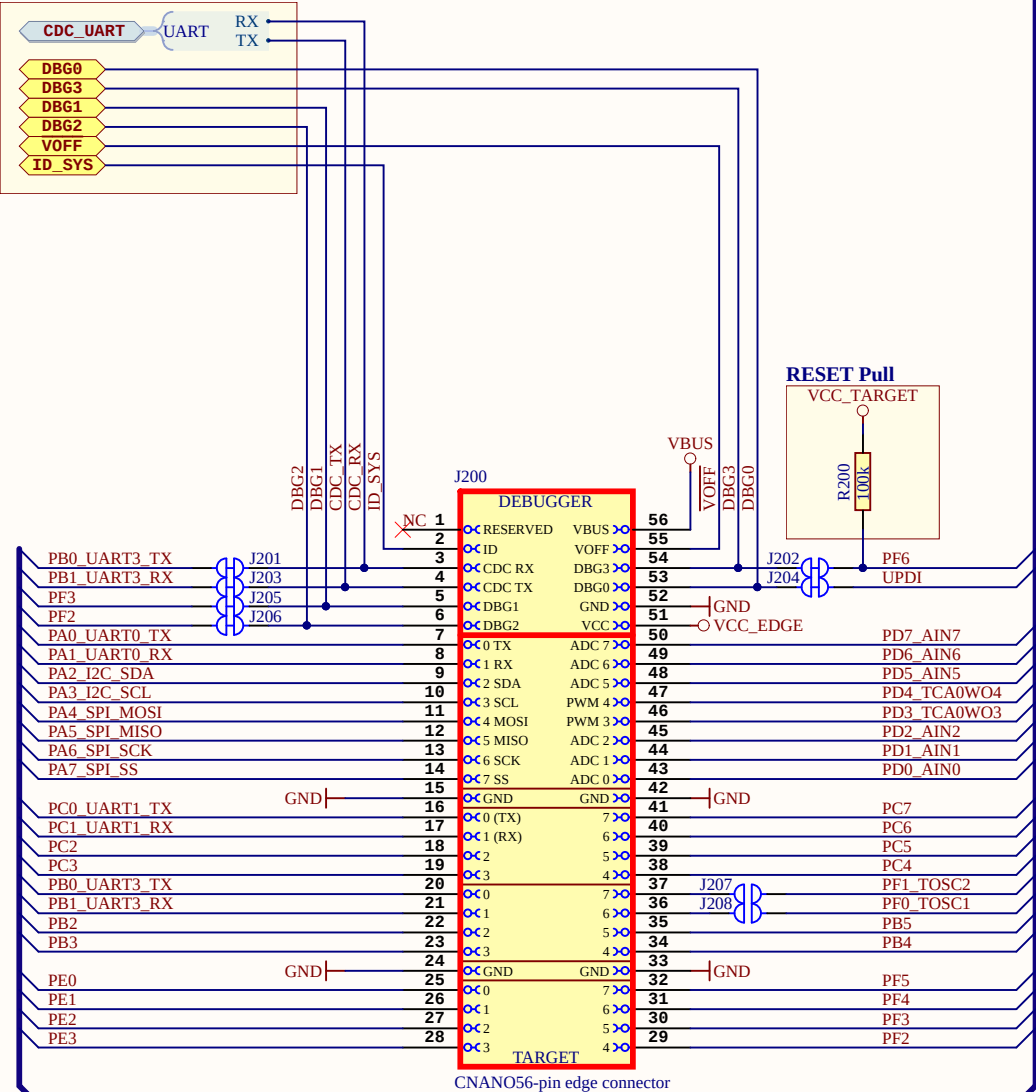
ATmega4809 Curiosity Nano



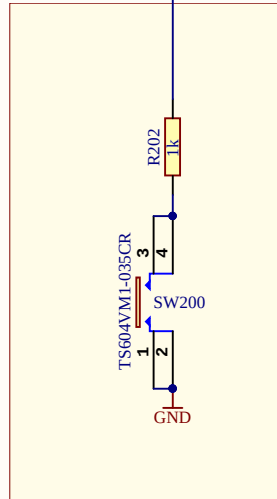
ATmega4809

ATmega4809		
Debugger	Name	Pin
CDC TX	UART3 RX	PB1
CDC RX	UART3 TX	PB0
DBG0	UPDI	UPDI1
DBG1	GPIO1	PF3
DBG2	GPIO0	PF2
DBG3	RESET	PF6
VTG	1.8V - 5.5V	

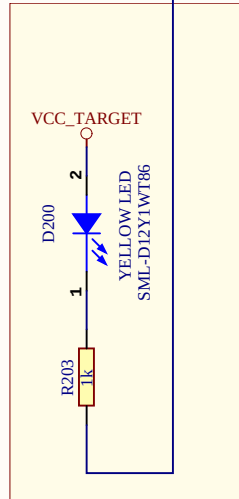
DEBUGGER CONNECTIONS



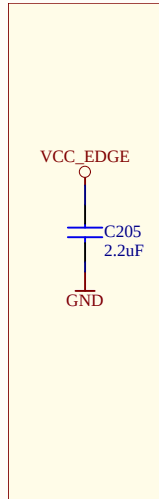
USER BUTTON



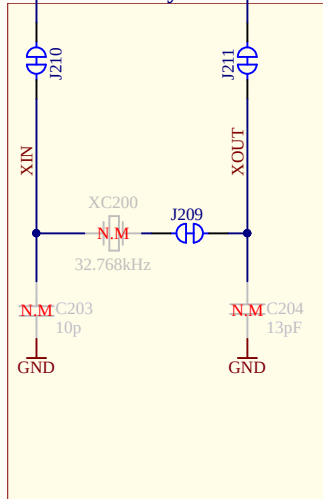
USER LED



TARGET BULK



32kHz Crystal



NOTE on Crystal calculations:

The calculations are based on a crystal used on an earlier revision of this board.

Crystal datasheet:
Ccrystal = 7pF
max ESR = 70kOhm
Accuracy ±20ppm
ATmega4809 datasheet:
Cxin = 5.5pF
Cxout = 5.5pF
 $C1 \approx 1 / ((1/5.5pF) + (1/5.5pF)) \approx 2.75pF$
Maximum Load = 12.5pF
Maximum ESR = 80kOhm

Estimated Cpcb = 0.5pF

Estimated load
 $C = 2 (Ccrystal - Cpara - Cpcb)$
 $C = 2 (7pF - 2.75pF - 0.5pF)$
 $C = 7.5pF$

Selected in design after verification
 $C = 10/13pF$

NOTE on UART/CDC:

RX/TX on the header denotes the input/output direction of the signal respective to it's source.

CDC TX is output from the DEBUGGER.
CDC RX is input to the DEBUGGER.
TX is output from the TARGET device.
RX is input to the TARGET device.

NOTE on I2C:

No pull-ups on board. Pull-ups should be mounted close to slave device(s).

Drawn By:
Microchip Norway
Engineer:
HN/TF



Project Title
ATmega4809 Curiosity Nano
Sheet Title
Target MCU

Size A3 PCB Assembly Number: A09-3094

PCB Number: A08-2838

File: ATmega4809_Curiosity_Nano_Target_MCU.SchDoc

PCBA Revision: 8

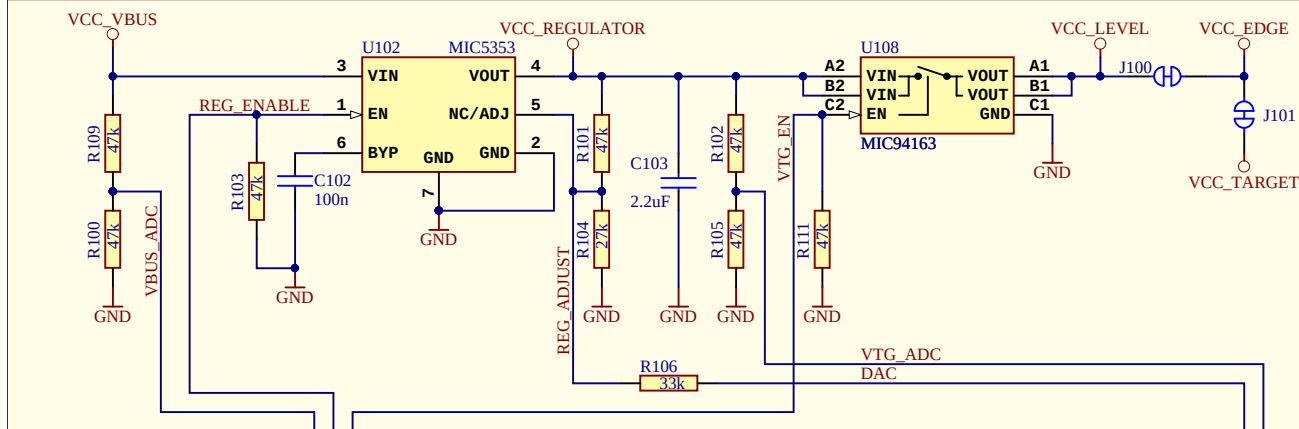
PCB Revision: 7

Date: 11/8/2019

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Page: 2 of 4

TARGET ADJUSTABLE REGULATOR



Adjustable output and limitations:

- The DEBUGGER can adjust the output voltage of the regulator between 1.25V and 5.1V to the target.
- The level shifters have a minimal voltage level of 1.65V and will limit the minimum operating voltage allowed for the target to still allow communication.
- The output switch has a minimal voltage level of 1.70V and will limit the minimum voltage delivered to the target.
- Firmware configuration will limit the voltage range to be within the target specification.
- Firmware feedback loop will adjust the output voltage accuracy to within 0.5%.

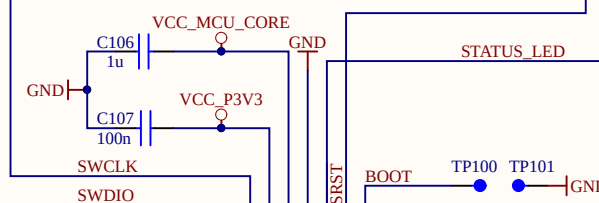
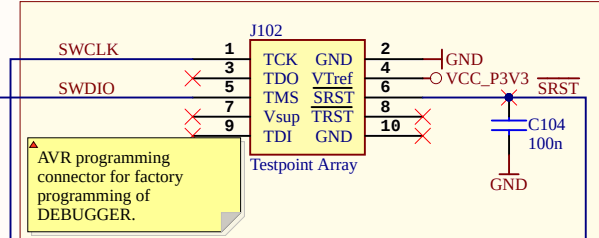
J100:
Cut-strap used for full separation of target power from the level shifters and on-board regulators.
- For current measurements using an external power supply, this strap could be cut for more accurate measurements. Leakage back through the switch is in the micro ampere range.

J101:
This is footprint for a 1x2 100mil pitch pin-header that can be used for easy current measurement to the target microcontroller and the LED / Button. To use the footprint:
- Cut the track between the holes, and mount a pin-header

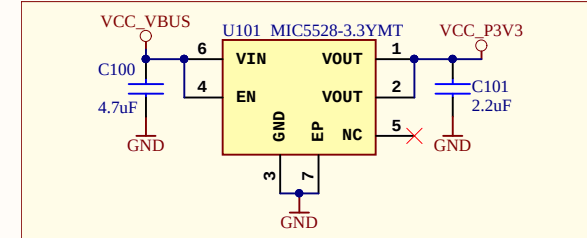
MIC5353:
Vin: 2.6V to 6V
Vout: 1.25V to 5.1V
Imax: 500mA
Dropout (typical): 50mV@150mA, 160mV @ 500mA
Accuracy: 2% initial
Thermal shutdown and current limit

Maximum output voltage is limited by the input voltage and the dropout voltage in the regulator.
(Vmax = Vin - dropout)

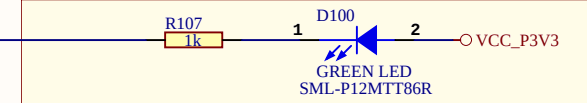
DEBUGGER TESTPOINT



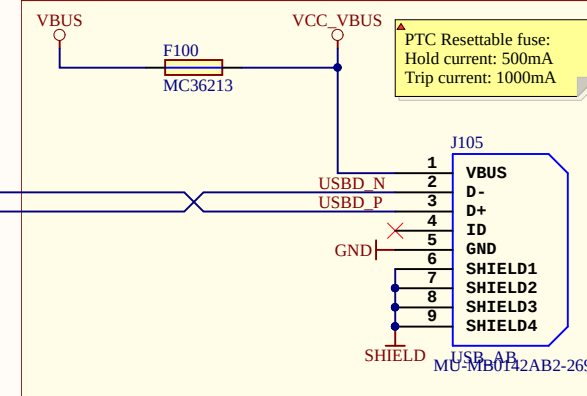
DEBUGGER REGULATOR



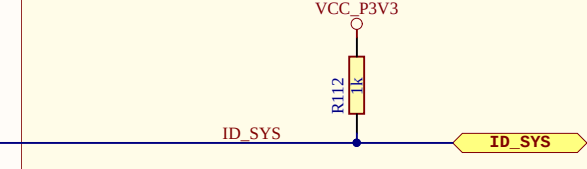
DEBUGGER POWER/STATUS LED



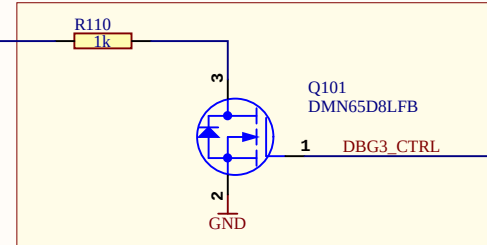
DEBUGGER USB MICRO-B CONNECTOR



ID PIN



DBG3 OPEN DRAIN



Interface	ICSP TARGET	UPDI TARGET
CDC TX	UART RX	UART RX
CDC RX	UART TX	UART TX
DBG0	DAT	UPDI
DBG1	CLK	GPIO
DBG2	GPIO	GPIO
DBG3	MCLR	RESET
VCC	-	-

MIC5528:
Vin: 2.5V to 5.5V
Vout: Fixed 3.3V
Imax: 500mA
Dropout: 260mV @ 500mA

Drawn By:
Microchip Norway
Engineer:
HN/TF



Project Title
ATmega4809 Curiosity Nano

Sheet Title
Debugger

Size A3 PCB Assembly Number: A09-3094

PCB Number: A08-2838

File: ATmega4809_Curiosity_Nano_DEBUGGER.SchDoc

PCBA Revision: 8

PCB Revision: 7

Date: 11/8/2019

Page: 3 of 4

Designed with
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Revision History

PCB Assembly Rev 1:

▲

Design Changes:

Initial Design

PCB:

PCB revision 1

PCB Assembly Rev 2:

▲

Design Changes:

Removed bypass switch

PCB:

PCB Revision 2, changed layout

PCB Assembly Rev 3:

▲

Design Changes:

Added reverse blocking switch (U108)
Added ID_SYS pin
Added VOFF pin
Change 10uf to 4.7uF (U100)

PCB:

PCB Revision 3
Increased board size for ID and VOFF pin
Rerouted for target voltage blocking switch

PCB Assembly Rev 4:

▲

Design Changes:

Added USB VBUS protection (U109)
Added VBUS pin

PCB:

PCB Revision 4
Increased board size for VBUS pin
Reroute the debugger section for the VBUS pin

PCB Assembly Rev 5:

▲

Design Changes:

Added input cap on U109

PCB:

PCB Revision 5
Added input cap on U109

PCB Assembly Rev 6:

▲

Design Changes:

Changed to Cut-Straps with thru-holes in Debugger Section.
Replaced Limit Switch with Fuse. Changed VBUS and VCC_VBUS nets back to previous scheme on DEBUGGER schematic. Added cut-straps to crystal

PCB:

PCB Revision 6
Changed layout to add Fuse and Cut-Straps with holes.
Adusted level-shift routing and adjusted Q101 footprint for improved assembly quality.

PCB Assembly Rev 7:

▲

Design Changes:

Added solderstrap jumper with holes between edge connector power (VCC) and VCC_TARGET.

PCB:

PCB Revision 7
Updated J200 (edge pin header connector) with staggered footprint, and adjusted traks, polygons, teardrops, and text accordingly.

PCB Assembly Rev 8:

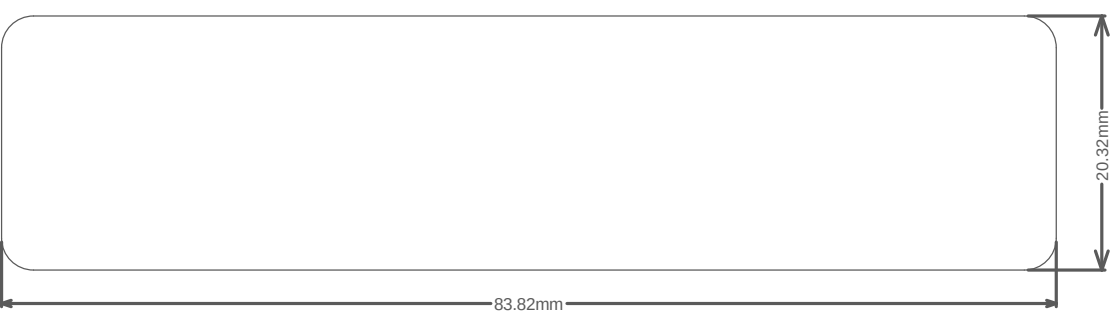
▲

Design Changes:

Replaced obsolete 32.768kHz crystal XC200.

PCB:

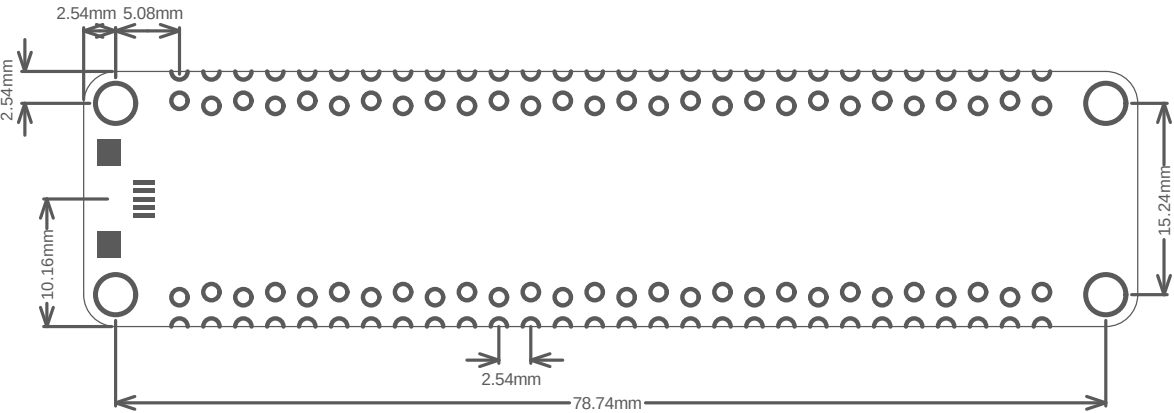
PCB Revision 7
No changes to PCB, stays at rev7.



83.82mm

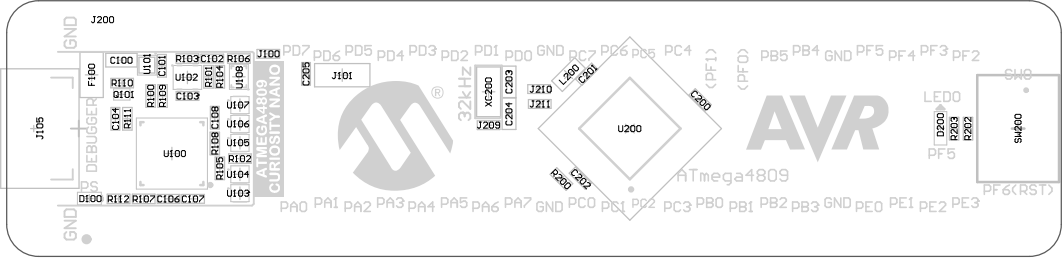
20.32mm

Connector Placement



Test Point Placement





A08-2838 Rev7

Diagram showing the connection of the PFI0 and PFI1 pins of the J208 and J207 connectors. The PFI0 pin of J208 is connected to the PFI0 pin of J207. The PFI1 pin of J208 is connected to the PFI1 pin of J207. The diagram also shows the PC4, PC5, PC6, PC7, and GND pins of the J208 connector.

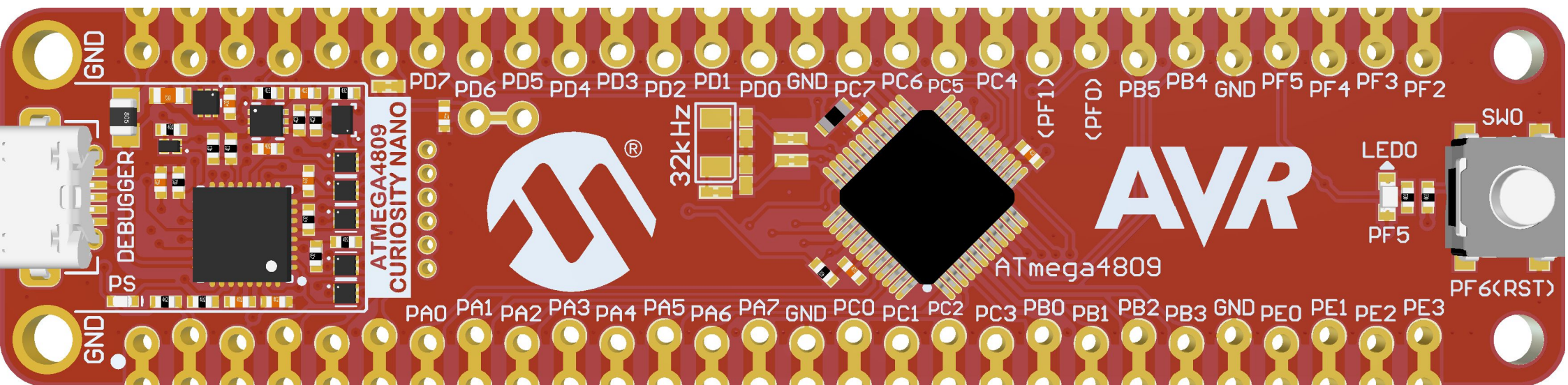
CONNECTIONS		
PF6	J202	D3
PF2	J206	D2
PF3	J205	D1
UPDI	J204	D0
PB0	J201	RX
PB1	J203	TX

TP101

TP100

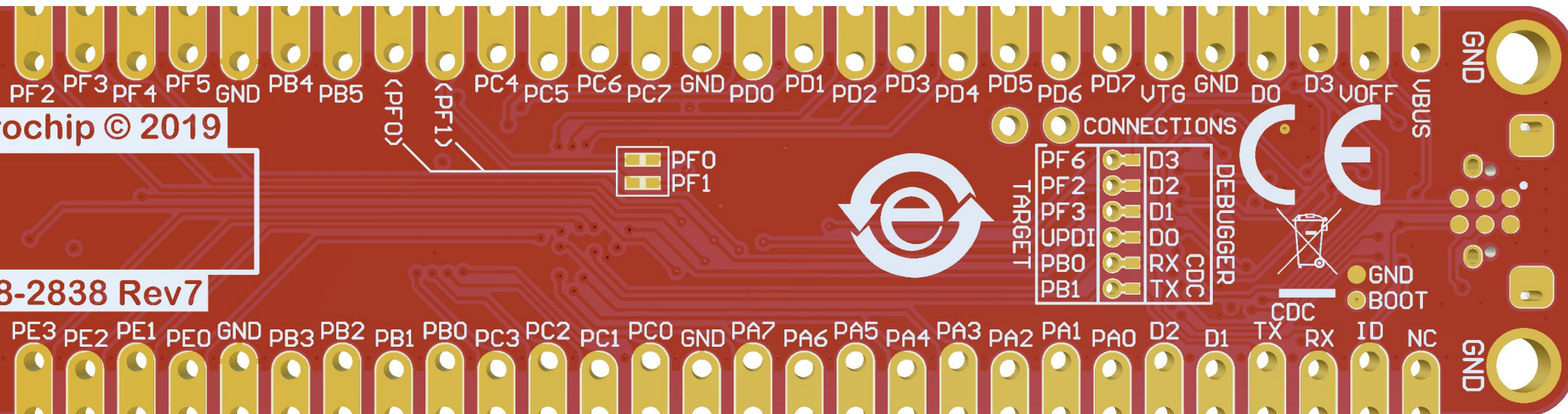
J102

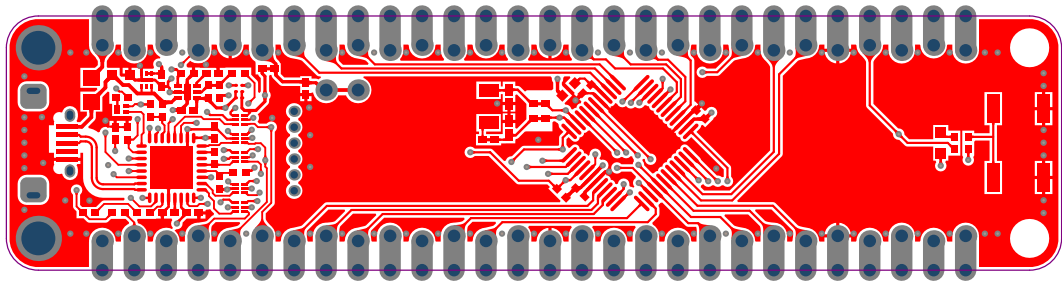
END

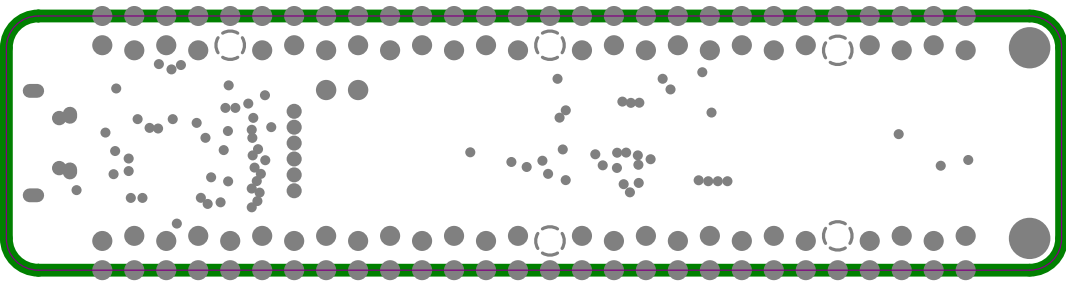


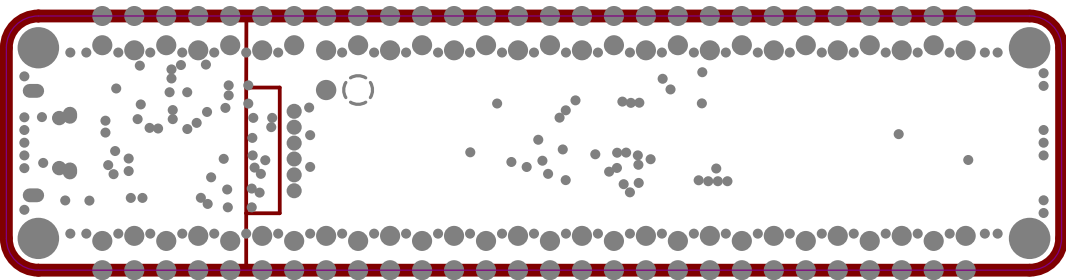
Microchip © 2019

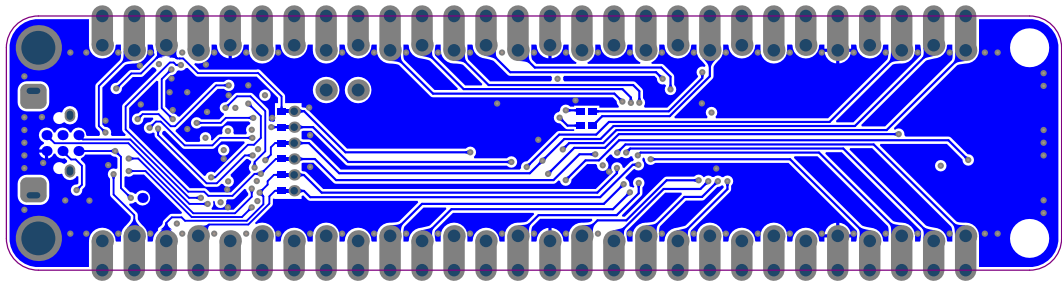
A08-2838 Rev7











Component list

Bill of Materials Fitted for Variant [Default Assembly] of Project [ATmega4809_Curiosity_Nano.PrjPcb] (No PCB Document Selected)

Source Data From: ATmega4809_Curiosity_Nano.PrjPcb
 Project: ATmega4809_Curiosity_Nano.PrjPcb
 Variant: Default Assembly



Report Date: 11/8/2019 2:51 PM
 Print Date: 11/8/2019 2:30:53 PM

Fitted	Designator	Quantity	Value	Manufacturer	MPN	Description
Fitted	C100	1	4.7uF	WALSIN Technology Corporation	0603X475K100CT	Ceramic capacitor, SMD 0603, X5R, 10V, 10% (de31036)
Fitted	C101	1	2.2uF	Kemet	C0402C225M9PAC	Ceramic capacitor, SMD 0402, X5R, 6.3V, +/-20%
Fitted	C102, C104, C107, C108, C200, C201, C202	7	100n	Kemet	C0402C104K4RACTU	Ceramic capacitor, SMD 0402, X7R, 16V, +/-10%
Fitted	C103, C205	2	2.2uF	TDK	C1005X5R1A225K	CAP CER 2.2UF 10V 10% X5R 0402
Fitted	C106	1	1u	Kemet	C0402C105K9PAC	Ceramic capacitor, SMD 0402, X5R, 6.3V, +/-10% (de26942)
Fitted	D100	1	GREEN LED	ROHM	SML-P12MTT86R	LED, SMD 0402, Green, Wave length=569nm, 2.1mcd @ (1mA, 1.9Vf)rohm
Fitted	D200	1	YELLOW LED	ROHM	SML-D12Y1WT86	LED, SMD 0603, Yellow, Wave length=590nm, 100mcd @ (20mA, 2.2Vf) rohm
Fitted	F100	1	MC36213	Multicomp	MC36213	Resettable PTC fuse, Ih = 0.5A, It = 1.0A, 0805 package
Fitted	FW1	1	nEDBG firmware			nEDBG firmware
Fitted	J105	1	MU-MB0142AB2-269	Allen Creations Corp.	MU-MB0142AB2-269	USB micro AB, Surface mount signals and DIP shield
Fitted	L200	1	BLM18PG471SN1	Murata	BLM18PG471SN1	SMD RF inductor 0603. Z=470Ohm (@100MHz), Max R(dc)=0.20Ohm, Max current=1A
Fitted	LABEL1	1	Label PCBA	ACT Logimark AS	505462	PCBA identification label PP Top White Gloss
Fitted	PCB1	1	ATmega4809 Curiosity Nano PCB documentation			ATmega4809 Curiosity Nano PCB documentation
Fitted	PCBADOC1	1	A09-3094 PCBA files			ATmega4809 Curiosity Nano PCBA documentation
Fitted	Q101	1	DMN65D8LFB	Diodes Incorporated	DMN65D8LFB-7	N-channel MOSFET, DFN1006-3 (SOT883), 60V, 330mA, 4Ohm
Fitted	R100, R101, R102, R103, R105, R109, R111	7	47k	KOA	RK73H1ETTP4702F	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	R104	1	27k	Yageo	RC0402FR-0727KL	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	R106	1	33k	ASJ Holdings	CR10-3302-FK	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	R107, R108, R110, R112, R202, R203	6	1k	ASJ Holdings	CR10-1001-FK	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	R200	1	100k	ASJ Holdings	CR10-1003-FK	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	SW200	1	TS604VM1-035CR	Dailywell Electronics Co.LTD	TS604VM1-035CR-R	SWITCH, SMD, 260gf, 6.4mm X 6.2mm
Fitted	TEST1	1	ATmega4809 Curiosity Nano test			Fixture test for ATmega4809 Curiosity Nano
Fitted	TESTDOC1	1	Curiosity Nano Test Instructions			Generic Test Instructions for Curiosity Nano
Fitted	U100	1	SAMD21E18A-MUT	Microchip	ATSAMD21E18A-MUT	32-bit RISC MCU 32pin
Fitted	U101	1	MIC5528-3.3YMT	Microchip	MIC5528-3.3YMT-T5	LDO 3.3V 0.5A 6TDFN
Fitted	U102	1	MIC5353	Microchip	MIC5353YMT-TR	500mA Ultra Low Dropout LDO regulator, 2% accuracy, 1.6x1.6mm MLF
Fitted	U103, U104, U105, U106, U107	5	74LVC1T45FW4-7	Diodes Incorporated	74LVC1T45FW4-7	Single-Bit Dual-Supply Transceiver, 1.65-5.5 Translation and 3-State Outputs
Fitted	U108	1	MIC94163	Microchip	MIC94163YCS-TR	Loadswitch, Rds(on) = 14.5mohm, 1.0mm x 1.5mm WLCSP, reverse blocking
Fitted	U200	1	ATmega4809-AFR	Microchip	ATmega4809-AFR	Atmel 8-bit RISC MCU, 48K Flash, 48pin TQFP, 7x7mm, 0.5m pitch package
Not Fitted	C203	0	10p	AVX	04025A100JAT2A	Ceramic capacitor, SMD 0402, NP0, 50V, +/-5%
Not Fitted	C204	0	13pF	Murata	GRM1555C1H130JA01D	Ceramic capacitor, SMD 0402, NP0, 50V, +/-5%
Not Fitted	XC200	0	32.768kHz	Abracon	ABS07-32.768kHz-7-T	Crystal, 32.768kHz, CL=7.0pF, ESR=70kOhm, SMD LxW=3.2 x 1.5mm, 20ppm
		51				

Approved	Notes